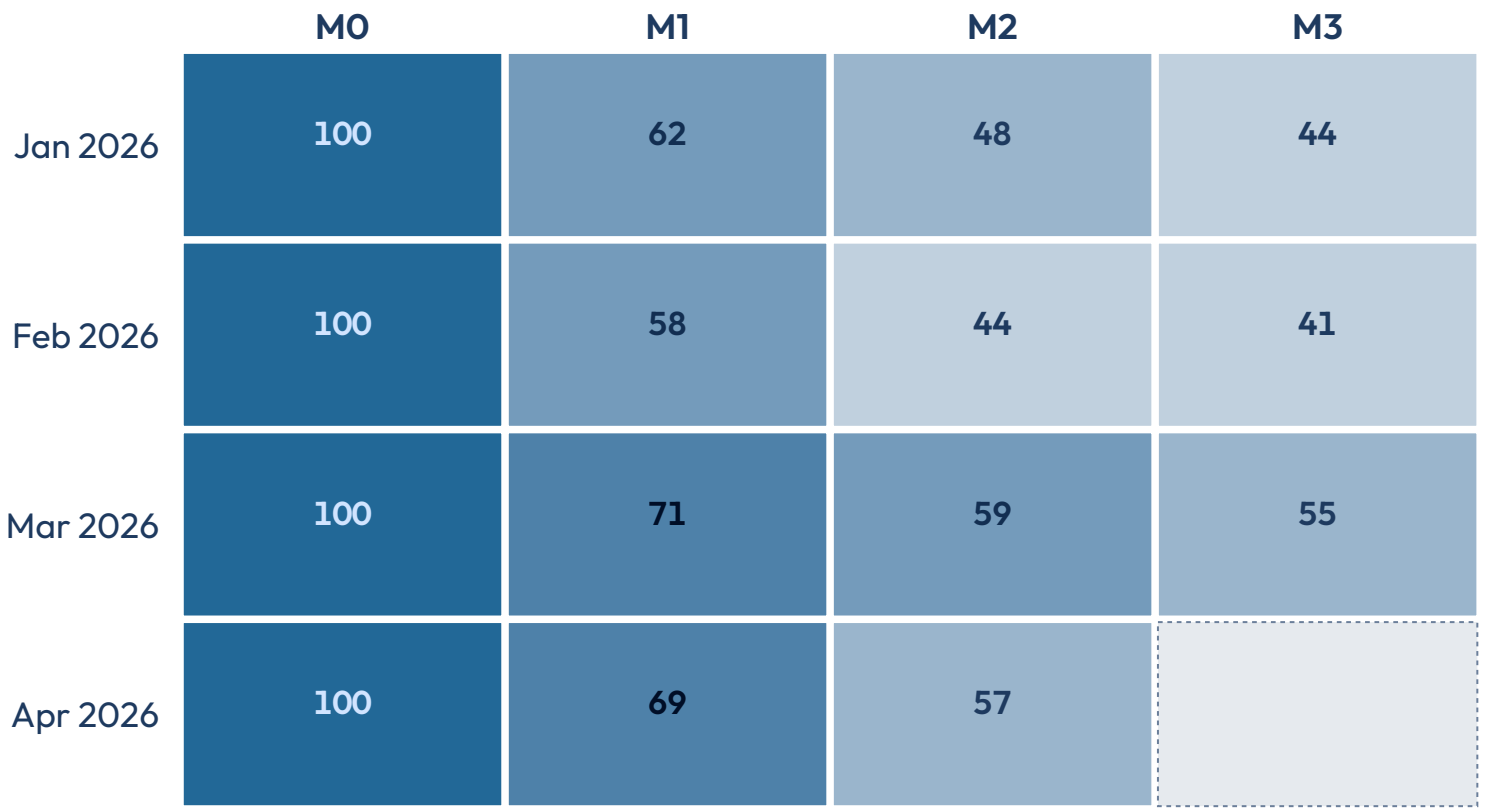
HEATMAP · A NUMERIC MATRIX READ AS INTENSITY

Where does it concentrate?

The chart family could compare magnitudes, decompose a total, plot a trend and score a 2×2 — but it had no way to put a **number at every crossing of two dimensions**. Cohort retention, a risk matrix, usage by hour and weekday: the shape a deck reaches for when the finding is a pattern rather than any single value.

RETENTION · 2026 COHORTS

Retention decays fastest in month two.



Four cohorts, four months — the shape first, the number on it

A missing cell is not a zero.

April has no month-three number because April has not reached month three. That is the **normal** state of a cohort table, not an authoring error — so the crossing paints the bare neutral with a dotted edge and carries no intensity at all. Pad it with `0` instead and the chart states a measurement nobody took, at the bottom of the ramp, where it reads as a collapse.

The ramp is the choropleth's, not a new one.

A heatmap and a `map` encode the same thing — one continuous value as one intensity — differing only in whether the cells are a grid or a geography. So this takes the mechanism `map` already proved: one hue mixed into a **neutral anchor**, never into the page. Mixing toward the canvas looks identical on light and inverts on dark, where a low value sinks *below* the empty cell it is supposed to sit above.

The ramp is five stops, so the number can sit on it.

Text on a colored fill is the one placement this family will not otherwise guarantee. The first cut printed the value on a **continuous** ramp and flipped its ink at one crossover — measured across all **33 palettes**, that leaves **318 cells** below AA, worst `2.60:1`. The fix is to stop asking for an exception: quantize. Five chosen stops means five fills to solve an ink against, and we choose which five — so the ramp steps *over* the band where no ink of any color reaches `4.5:1`. Each theme carries its own five inks, solved against the fill each stop actually paints.

RETENTION · THE CEILING

Ten cohorts by twelve months.

	M0	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11
Jan 2026	100	71	59	55	52	50	48	47	46	45	44	43
Feb 2026	100	58	44	41	39	38	37	36	35	34	33	
Mar 2026	100	62	48	45	43	42	41	40	39	38		
Apr 2026	100	69	57	54	51	50	49	48	47			
May 2026	100	66	54	51	49	48	47	46				
Jun 2026	100	64	51	48	46	45	44					
Jul 2026	100	73	61	58	56	55						
Aug 2026	100	60	47	44	42							
Sep 2026	100	68	56	53								
Oct 2026	100	75	64									

When not to reach for it.

Intensity is read approximately — that is its strength for a pattern and its weakness for a comparison. One row of numbers is not a matrix; it is a `bar`, where length is judged precisely. Cells that are verbs or owners belong in `matrix-grid`. A value per country belongs on `map`. And if the decision turns on which of two similar cells is larger, the value belongs on an axis.